

GenAI-Powered Customer Feedback Insights Assistant

Capstone Project Problem Statement

Project type	Individual capstone project
Business context	Customer feedback analysis for a growing subscription-style business
Primary skill focus	Generative AI, LLM use, prompt engineering, feedback analysis, business recommendations, and portfolio-ready storytelling
Dataset	UCI Sentiment Labelled Sentences Dataset — use Amazon and Yelp review files for the main analysis; IMDb is optional for extension.
Final output	A concise business-ready submission with insights, reusable prompts, a simple GenAI workflow/prototype, and a 6–8 slide final presentation.
Expected effort	Approximately 12 hours total

1. Common Problem Statement

A growing business receives a large volume of customer reviews and feedback, but the team does not have a fast or consistent way to identify major pain points, sentiment patterns, and improvement opportunities.

The leadership team wants to know: What are the top customer pain points, what sentiment patterns are visible, and what actions should the business prioritize to improve customer satisfaction?

Your task is to use Generative AI to analyze the provided review dataset, identify recurring themes, explain positive and negative sentiment patterns, create reusable prompts, and present clear business recommendations.

2. Business Scenario

Assume you are working as an AI-enabled business analyst supporting the Head of Customer Experience. The business wants to convert customer feedback into actionable insights quickly and repeatedly.

- The business wants to understand what customers like and dislike.
- The leadership team wants to prioritize the most important improvement areas.
- The customer experience team wants a reusable GenAI workflow that can be repeated for future feedback batches.
- Your final submission should help a business stakeholder move from raw feedback to decisions.

3. Dataset to Use

Use the UCI Machine Learning Repository’s Sentiment Labelled Sentences Dataset. The dataset contains 3,000 text sentences labelled as positive or negative sentiment. It includes review sentences from Amazon, Yelp, and IMDb, with each source containing 500 positive and 500 negative sentences. The label is 1 for positive sentiment and 0 for negative sentiment.

For this capstone, use the Amazon and Yelp files as customer feedback/review data. IMDb reviews may be used only as an optional extension.

Dataset Files

File	Recommended use	Records	Notes
amazon_cells_labelled.txt	Main analysis	1,000	Product/customer review style feedback
yelp_labelled.txt	Main analysis	1,000	Service/restaurant review style feedback
imdb_labelled.txt	Optional extension	1,000	Use only if you want a larger or comparison dataset

File	Recommended use	Records	Notes
readme.txt	Reference	-	Contains dataset notes

Dataset page: [UCI Sentiment Labelled Sentences Dataset](#) | Citation: Kotzias, D. (2015). Sentiment Labelled Sentences [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C57604.

4. How to Download the Dataset

Access note

The UCI dataset is freely available and does not require a paid subscription. If the website design changes slightly, search for “UCI Sentiment Labelled Sentences dataset” and use the official UCI Machine Learning Repository page.

1. Open the official UCI dataset page: UCI Sentiment Labelled Sentences Dataset.
2. Click the Download button on the dataset page. The downloaded file will be a ZIP file.
3. Unzip the downloaded file on your computer.
4. Open the folder named sentiment labelled sentences.
5. Locate the files amazon_cells_labelled.txt and yelp_labelled.txt.
6. Use these two files for your main capstone analysis. The imdb_labelled.txt file is optional.
7. Each row contains a review sentence followed by a sentiment label: 1 = positive, 0 = negative.

Optional: Combine the Amazon and Yelp Files

You may analyze the files separately or combine them into one working file with the following columns. Optional Python helper code is provided in the appendix.

Column	Description	Example
feedback_text	The customer review sentence	The service was slow and the food was cold.
sentiment_label	Original dataset label	0
source	Dataset source	Yelp

5. Your Capstone Task

Build a GenAI-powered Customer Feedback Insights Assistant. The assistant does not need to be a production application. It can be a prompt workflow, spreadsheet workflow, Python notebook, or simple prototype. It must show how raw review text can be converted into useful business insights and recommendations.

Minimum Capstone Requirements

- Analyze customer review text from the Amazon and Yelp files.
- Identify the top recurring customer pain points and positive themes.
- Explain sentiment patterns using the original labels and your AI-assisted interpretation.
- Create a reusable prompt library with at least 6 prompts.
- Produce a simple GenAI workflow/prototype that demonstrates how the analysis can be repeated.
- Present findings and recommendations in a clear business-ready format.

6. Required Business Questions

1. What are the top recurring customer pain points?
2. Which feedback themes are most associated with negative sentiment?
3. What positive patterns should the business preserve or strengthen?
4. What are the likely root causes behind the most common complaints?
5. What actions should the business prioritize to improve customer satisfaction?
6. How can GenAI automate this type of customer feedback analysis in the future?

7. Suggested Analysis Approach

Step	What to do	Expected output
1. Inspect	Review the dataset structure, file sources, row counts, and sentiment labels.	Dataset overview
2. Prepare	Clean text if needed, combine Amazon and Yelp files, and create a source column.	Working dataset
3. Explore	Use GenAI and/or simple counts to identify frequent complaint and praise themes.	Initial theme list
4. Prompt	Design reusable prompts for summarization, theme extraction, root-cause analysis, and recommendations.	Prompt library
5. Validate	Check whether AI-generated themes make sense by reviewing sample rows from each theme.	Quality notes and corrections
6. Recommend	Translate findings into specific business actions with expected impact.	Recommendations table
7. Present	Create a short business-facing presentation.	6–8 slide final presentation

8. Required Deliverables

- 1. Dataset Summary:** Briefly describe the files used, number of rows analyzed, sentiment labels, and any cleaning or preparation done.
- 2. Customer Feedback Insights Summary:** Provide the top findings using evidence from the dataset and explain what they mean for the business.
- 3. Reusable Prompt Library:** Submit at least 6 reusable prompts covering summarization, theme extraction, sentiment explanation, root-cause analysis, recommendation generation, and executive summary writing.
- 4. GenAI Workflow or Prototype:** Show how your solution takes customer feedback as input and produces insights or recommendations as output. This may be a prompt workflow, spreadsheet, notebook, script, or simple assistant design.
- 5. Final Presentation:** Create a 6–8 slide presentation explaining the problem, dataset, workflow, prompt strategy, findings, recommendations, limitations, and next steps.
- 6. Reflection:** Briefly explain where GenAI helped, where it may have produced weak or risky outputs, and how you checked or improved the result.

9. Required Prompt Library

Prompt type	Purpose	Minimum expectation
Feedback summarization	Summarize a batch of reviews	Clear summary of positive and negative feedback
Theme extraction	Identify recurring pain points or praise themes	Structured output with theme names and examples
Sentiment explanation	Explain why feedback appears positive or negative	Ground explanation in the review text
Root-cause analysis	Infer likely business causes behind complaints	Separate evidence from assumptions
Recommendation generation	Convert findings into practical actions	Prioritized actions with rationale
Executive summary	Write a stakeholder-ready summary	Concise business language with next steps

Example Prompt

You are a customer experience analyst. Analyze the following customer review records. Identify the top recurring themes, separate positive and negative themes, provide 2–3 evidence examples for each theme, and recommend one business action per negative theme. Format the output as a table with columns: Theme, Sentiment Direction, Evidence, Business Meaning, Recommended Action.

10. Project Plan

Phase	Activity	Output
1	Understand the problem statement and dataset	Project understanding notes
2	Download, load, and review the UCI dataset	Dataset summary
3	Clean and structure review text	Working analysis file
4	GenAI-assisted theme and sentiment analysis	Initial findings
5	Prompt library creation and iteration	At least 6 reusable prompts
6	Build simple GenAI	Workflow/prototype evidence

	workflow/prototype	
7	Insights, recommendations, and validation	Final findings table
8	Final presentation and submission packaging	Final capstone submission
	Total	



11. Final Presentation Structure

Recommended length: 6–8 slides.

- Slide 1: Project title and business problem
- Slide 2: Dataset overview and analysis scope
- Slide 3: GenAI workflow/prototype
- Slide 4: Prompt strategy and examples
- Slide 5: Key findings and evidence
- Slide 6: Recommended business actions
- Slide 7: Risks, limitations, and validation approach
- Slide 8: Next steps and how the assistant could be improved

12. Assessment Rubric

Criteria	Weight	What strong work looks like
Understanding of the business problem	10%	Frames the customer feedback challenge clearly and connects analysis to business decisions.
Dataset review and preparation	15%	Uses the correct public dataset, prepares it appropriately, and explains scope and limitations.
Quality of GenAI-assisted analysis	20%	Produces meaningful themes, sentiment insights, and evidence-based findings.
Prompt design and iteration	20%	Prompts are reusable, structured, specific, and improved through testing.
Business recommendations	15%	Recommendations are practical, prioritized, and linked to findings.
Workflow/prototype clarity	10%	Workflow clearly shows how feedback becomes insights and recommendations.
Final presentation and storytelling	10%	Submission is concise, professional, and easy for a stakeholder to understand.

13. Scope Control

Keep in scope	Avoid
Amazon and Yelp review files	Large external datasets
Theme and sentiment analysis	Building a production-grade app
At least 6 reusable prompts	Dozens of prompts without testing
Simple workflow/prototype	Complex model training
6–8 slide final presentation	Long report with unnecessary detail
Evidence-based recommendations	Recommendations not tied to review data

14. Final Submission Checklist

- ☐ Dataset summary
- ☐ Customer feedback insights summary
- ☐ Reusable prompt library with at least 6 prompts
- ☐ GenAI workflow/prototype evidence
- ☐ Final 6–8 slide presentation
- ☐ Reflection on GenAI limitations and validation

15. Dataset Source and Use

Dataset source: [UCI Sentiment Labelled Sentences Dataset](#). The UCI page lists the dataset as licensed under Creative Commons Attribution 4.0 International (CC BY 4.0), which allows sharing and adaptation with appropriate credit.

Learners should cite the dataset in their final submission as: Kotzias, D. (2015). Sentiment Labelled Sentences [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C57604.

Appendix: Optional Python Helper Code

Use this only if you want to combine the Amazon and Yelp text files into one CSV file. Adjust the folder path if needed.

```
import pandas as pd

files = [
    ("Amazon", "amazon_cells_labelled.txt"),
    ("Yelp", "yelp_labelled.txt")
]

frames = []
for source, filename in files:
    df = pd.read_csv(f"sentiment labelled sentences/{filename}", sep="\t",
                    header=None, names=["feedback_text", "sentiment_label"])
    df["source"] = source
    frames.append(df)

feedback_df = pd.concat(frames, ignore_index=True)
feedback_df.to_csv("capstone_customer_feedback.csv", index=False)
```

